Using Small-Area Estimation (SAE) to Estimate Prevalence of Child Health Outcomes at the Census Regional-, State-, and County-Levels

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Abstract

In this study, we implement small-area estimation to assess the prevalence of child health outcomes at the county, state, and regional levels, using national survey data.

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[1] Frequent acronyms include CI: Confidence Interval, CV RMSPE: cross-validation based on the root mean squared predictive error, FIPS: Federal Information Processing Standard, FSRDC: Federal Statistical Research Data Center, ICAR: intrinsic conditional autoregressive, NSCH: National Survey of Children's Health, RDC: Research Data Center, SAE: Small Area Estimation, TS: Tourette Syndrome, and UofSC: University of South Carolina

Small-Area Estimation (SAE): Importance and Utility

Prevalence estimates are critical sources of information that inform policy and program development, indicating for whom and where efforts are most needed. Small-area estimation (SAE), in particular, is a previously used and well-established method to generate prevalence estimates (Battese et al., 1988; Ghosh & Rao, 1994), as well as associated confidence intervals. Multilevel model-based SAE is useful when accuracy of more-underrepresented or smaller communities is important. Specifically, large population-based surveys, such as those gathered by the Census Bureau (e.g., National Survey of Children's Health (NSCH)), are well powered to provide direct estimates nationally or at the state-level but cannot provide estimates for smaller areas due to small sample sizes at the local level (e.g., Census tracts, county; Myers et al., 2015; Panczak et al., 2016; Zhang, 2013; Zhang et al., 2015). SAE then harnesses data from the larger areas to strengthen estimates for the smaller areas. These smaller area estimates are of critical interest because they allow interventions to target more specific, local communities and help inform resource allocation decisions.

For clarity, SAE refers to the process of using data from areas with larger sample sizes to inform predictions for areas with smaller sample sizes, while multilevel refers to the utilization of several types of information. For instance, outcome data can be a child-level prevalence estimate for a county, state, or region (e.g., autism spectrum disorder, overweight/obesity, and attention-deficit/hyperactivity disorder; Bradshaw et al., under review; Zgodic et al., 2021, under review), while predictors can be child-level (e.g., race, ethnicity, gender, age, parental educational attainment) *and/or* area-level (e.g., regional division characteristics, federal/state/county school funding, county child insurance rates). Finally, model-based refers to statistical models used to generate the estimates[2]. Not only is SAE useful because of its ability to provide estimates for less-data rich areas, it also has a wide variety of health applications, such as obesity, chronic disease, vaccination status, HIV, chronic obstructive pulmonary disease, and health-related quality of life (Benjamin Neelon et al., 2017; Cataife, 2014; Eberth et al., 2013; Fang et al., 2018; Khan et al., 2020; Klompas et al., 2017; Koh et al., 2018; Li et al., 2016; Lin et al., 2017; Mills et al., 2020; Tabano et al., 2017; X. Zhang et al., 2014; Z. Zhang et al., 2011). This enables SAE to be a valuable approach to utilize with the NSCH dataset and similar datasets. Thus far, the research team at the University of South Carolina (UofSC) has utilized SAE to estimate Tourette Syndrome (TS; see section "Tourette Syndrome (TS)"), autism spectrum disorder (Bradshaw et al., under review), overweight/obesity (see section "Overweight and Obesity"; Zgodic et al., 2021), and attention-deficit/hyperactivity disorder (Zgodic et al., under review) prevalence estimates from the NSCH.

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[2] For instance, Zgodic et al. (2021) utilized a multilevel mixed effects logistic regression model (for child *j* in county *k*) with a county-level intrinsic conditional autoregressive random intercept term to predict county-level child overweight/obesity prevalence and used the following model ($\mathbf{X}_{jk}$ represented a child-level vector, $\mathbf{C}_k$ represented an area-level vector, and b_k represented spatial dependence between neighboring counties): $logit(p_{jk}) = \mathbf{X}_{jk}\boldsymbol{\beta}_{\boldsymbol{child}} + \boldsymbol{C}_k\boldsymbol{\beta}_{\boldsymbol{area}} + b_k$.

Approach and Methods: How to Utilize Restricted-Use and Publicly Available Data

To use SAE on a combined dataset of restricted-use and publicly available data, the research team at the University of South Carolina utilized the following process:

1. Applied to use restricted data
2. Gathered non-restricted data
3. Combined the data into a single dataset in the Research Data Center's (RDC's) secure environment
4. Accessed all data and conducted statistical analysis in the RDC's secure environment
5. Applied a variable selection procedure to the model in the RDC's secure environment
6. Worked with RDC staff to get estimates released from the secure environment (ensuring that respondent confidentiality was not compromised)
7. Generated small area estimates and confidence intervals (outside of RDC).

Restricted-Use Data

The research team at the University of South Carolina utilized the NSCH dataset, which is a national survey examining the emotional and physical health of children in the United States (U.S.), ages 0-17 years. The NSCH was sponsored by the Health Resources and Service Administration and conducted by the U.S. Census Bureau. An equal number of randomly selected homes per state (including the District of Columbia) were contacted, such that parents/guardians provided health information about one child in that home, who was also randomly selected; for the 2016-2020 datasets, data were collected by online survey or by mail. The NSCH gathered information regarding child physical and emotional health, family interactions, parental health, school experiences, and neighborhood safety.

To use restricted variables from the NSCH (e.g., county of residence), the research team underwent an extensive application process and met a set of specific requirements. Briefly, they completed an application with their Federal Statistical Research Data Center (FSRDC) Director and Census Administrator to use the restricted data, prepared specific documentation regarding details of their analyses and proposed project (e.g., background, methodology, timeline), and applied for Special Sworn Status (SSS) to access the restricted data. They also paid all necessary fees and coordinated visits to their RDC, as well as completed regular reports (e.g., annual, upon completion) and required trainings. Further, they submitted amendments when changes to the project were warranted.

Non-Restricted, Public-Use Data

To provide additional context and to strengthen their model building, the team at the UofSC also relied on public-use, secondary datasets. As this data was area-level (e.g., county, state, Census regional division), it was not restricted. The team utilized data regarding rurality, insurance coverage, school funding, state policies, and sociodemographic information, as well as community health, wellness, accessibility, and safety. This external data was extracted from governmental agencies and organizations, such as the Department of Health and Human

Services, Department of Agriculture, Environmental Protection Agency, U.S. Census Bureau, and American Community Survey.

Like with the restricted-use data, the research team also went through a process to use the public-use data. For each additional dataset they wanted to include in their analyses, they had to complete a User Provided Memo, where they provided information regarding the file type and user permissions. They also sent the requested public-use data to the RDC.

Combining Non-Restricted, Public-Use Data with Restricted Data

The research team requested to use public-use area-level, in addition to the restricted-use individual-level data from the NSCH dataset. Originally, these additional area-level variables were not linked to the individual-level data in the NSCH dataset, so the first step in the RDC's secure environment was to merge the area-level variables (see section "On-Site Analysis at the RDC") to the NSCH data using Federal Information Processing Standard (FIPS) codes, which are unique geographic identifiers. Both the NSCH and the public-use data have FIPS codes and allow each child (i.e., row) in the NSCH data to be merged with the county, state, and Census regional division data of the public-use data. Upon merging, the dataset was considered complete and ready for analysis. It is important to note that this completed, combined dataset now contained *many* variables. Later, the research team used a variable selection procedure to identify the most predictive set of variables (see section "Small Area Estimate Modeling on a Combined Dataset").

Analyses: On-Site at the RDC

To conduct the intended analyses, the research team's Principal Investigator (PI) traveled to the Triangle Federal Statistical Research Data Center (TRDC) in Durham, NC to conduct the analyses in the TRDC's secure computer lab (other RDCs can be found at https://www.census.gov/about/adrm/fsrdc/locations.html). The U.S. Census Bureau had previously vetted the requested restricted-use NSCH, such that the data was already in the RDC's secure environment upon arrival. Analyses occurred in SAS, a software, in a secure environment. The PI sent draft SAS code ahead of time as well, such that it could be edited accordingly (e.g., correct variable names) in the secure RDC environment.

At the TRDC, the workstations use a Red Hat Linux operating system, which is considerably different than the Windows operating system or macOS. Particularly, the SAS interface for Linux is not as user friendly as the interface on Windows. Researchers cannot connect to the internet on the workstations, mobile, or laptop. To connect to the internet, researchers must leave the RDC.

Small Area Estimate Modeling on a Combined Dataset

As noted in the introduction, SAE is a statistical method, such that areas with small sample sizes borrow information from areas with larger sample sizes, allowing for more representative prevalence estimates. The SAE models applied herein were all logistic mixed effects models:

$$logit(p_{jk}) = \boldsymbol{X}_{jk}\boldsymbol{\beta}_{\boldsymbol{child}} + \boldsymbol{C}_k\boldsymbol{\beta}_{\boldsymbol{area}} + b_k$$

where p_{jk} represents the probability of the outcome for child j in area k; $\boldsymbol{X}_{jk}$ represents the child-level fixed effects covariates (e.g., race/ethnicity, age, sex, parents' highest educational attainment); $\boldsymbol{C}_k$ represents the area-level fixed effects covariates; and b_k represents the random intercept for area k (e.g., county).

SAE can include spatial information as well, but it is not a requirement. To include spatial information, b_k was modeled using an intrinsic conditional autoregressive (ICAR; Banerjee et al., 2004; Cressie, 1993) model. As discussed in section "Results Release Process: Disclosure Review Meeting", this choice was particularly helpful in the disclosure process for the research team's models, as it made each county-level random effect prediction a function of the surrounding county-level random effect predictions. When b_k does not include spatial information, b_k is a traditional random intercept for a given area level, for example, the state level. This type of mixed effects model can be fitted using SAS PROC GLIMMIX. Sample code used in the "Tourette Syndrome (TS)") analysis can be found at: https://github.com/anjazgodic/ts_sae_poststratification.

The combined dataset included many predictor variables which could be used in the model. The next step, still in the RDC's secure environment, was variable selection. The research team opted to use 5-fold cross-validation based on the root mean squared predictive error (CV RMSPE), which minimizes the root mean squared predictive error (MSPE) of the model, because it favors variables that improve the predictive ability of the model. Specifically, the dataset was randomly split into five approximately equal groups (i.e., "folds"). Iteratively, *each fold* was used as a validation dataset, while the remaining four folds trained the model (Allen, 1974; Stone, 1974, 1977). The model fitted to the training set (i.e., the four folds) then was used to predict the prevalence for the desired outcome in the validation set (i.e., the remaining fifth fold) in which the RMSPE was estimated. This process was repeated five times, such that each fold acted as the validation set once, and the model producing the smallest RMSPE was selected as the final model. In doing so, only the most predictive, impactful covariates that minimized the RMSPE were maintained in the final model.

The researchers fitted the above logistic mixed effects model to the full dataset with the predictors identified in the variable selection process. Because SAE is a type of regression, the output from SAE analyses includes estimates of child- and area-level fixed effects (i.e., $\boldsymbol{\beta}_{\boldsymbol{child}}$ and $\boldsymbol{\beta}_{\boldsymbol{area}}$), predictions of the random effects (i.e., b_k's), the covariance matrix for the fixed effects, and the estimated variance of the predicted random effects. For example, if variable selection selected child-level and area-level predictors and if the model included a random area-level intercept, the model would produce child- and area-level fixed effects and a predicted random intercept for each area, along with their estimated standard errors. The SAE analyses generated effects that represent estimates that apply to *all* small and large areas (i.e., from fixed effects), as well as estimates specific to each area (i.e., from random effects). For example, one fixed effect could estimate the association of sex on obesity, which is constant across all areas. The random effects account for some of the variance in the outcome that is

unaccounted for by the fixed effects. The predicted random effects help make *predicted* area-level values (e.g., predicted proportion of obese children) closer to the *observed* area-level values (e.g., observed proportion of obese children). This happens such that the difference in predicted versus observed values is smaller for areas with larger sample sizes than for areas with smaller sample sizes (i.e., shrinkage estimation, see Breslow & Clayton, 1993).

Generating Results: Outside of the RDC

Results Release Process: Disclosure Review Meeting

After completing the analyses and generating fixed effects, random effects (for specific small areas, like counties), and their corresponding variances, the researchers submitted the appropriate documentation (see below) to their FSRDC Administrator for the release of the model coefficients *from* the RDC's secure environment. The FSRDC Administrator worked with disclosure avoidance staff in the Census Bureau's Center for Enterprise Dissemination, as well as Census's Disclosure Review Board (DRB), to examine the results, ensuring that the estimates abided by Census disclosure standards. The required documentation consists of all requested estimates, tables of sample sizes from which each estimate was derived (to ensure that sample sizes are appropriate), and an FSRDC Clearance Request Memo.[3]

Here, a Disclosure Review Board Meeting was required. At this meeting, members of the research team met with the FSRDC Administrator, disclosure avoidance staff, and the DRB to discuss the analyses and the risks of a potential disclosure violation. Because fixed effects are outcomes that use all predictors in the model, fixed effects are not often subject to very stringent reviews. However, some interaction terms may need to be removed due to them being based on small samples. The random effects are specific to small areas and undergo more scrutiny. County-level random effect estimates are (traditionally) only a function of people that were sampled from that county and would not pass the disclosure review. For the spatial ICAR model used in these analyses, the county-level random effect predictions are a function of children in that county, the children in the surrounding counties, the children in those counties that surround the surrounding counties, and so on. As a result, for the contiguous U.S., each county-level random effect prediction is a function of all children in the contiguous U.S. and pass disclosure review. Further, the county-level random effect predictions from Alaska pass disclosure review since the state is contiguous, and the predicted random effects are a function of all children from Alaska. For Hawaii, however, none of the counties share a common boarder; consequently, county-level random effect predictions cannot pass disclosure and were not requested for release. After the disclosure review meeting, the results (e.g., fixed effects, select random effects, and select random intercepts) were released to the researchers.

Poststratification: Making Representative Estimates from Population Counts

The fitted regression model from the SAE analysis results in prevalence estimates for all strata relevant to the variables (which were identified via variable selection (see section "Small Area Estimate Modeling on a Combined Dataset")) in the model for each small area. For instance, in

[3] Examples of the FSRCD Clearance Request Memo are available from the corresponding author upon request.

a model that uses race/ethnicity (four categories), biological sex (two categories), age (eight categories), and parental education (four categories), there are 256 different (256 = 4*2*8*4) possible combinations (strata) of demographic variables. Another model may use sex (two categories) and diagnostic status (four categories) as predictors, yielding eight strata of participant characteristics.

Poststratification

To make a single estimate for each small area, rather than as many estimates as strata per small area, the research team then used a technique called poststratification. Poststratification creates weighted averages that reflect a population's underlying distribution of characteristics. The research team multiplied the estimates for each stratum by the number of children per stratum in each small area (i.e., population counts) and divided the results by the number of children in that small area, resulting in a single small area estimate that reflected each small area's demographic make-up. Poststratification occurred outside of the RDC's secure environment.

Obtaining Population Counts

To conduct poststratification, the research team needed to obtain population counts for each stratum (e.g., the number of children per stratum in each small area). This can be accomplished through several mechanisms. Population counts may be available through public data (e.g., https://www.census.gov/topics/population.html). Alternatively, if population counts for individual-level factors are unavailable, researchers may have to generate population counts from additional SAE models (Bradshaw et al., under review) or multi-step processes (see section "Overweight and Obesity"). For example, population counts for adult education may be publicly available, but population counts for adult *parents* may be unavailable, and additional steps to obtain such counts may be required. Ultimately, predictors in the model dictate how the population counts should be obtained.

It should be noted that the use of an individual variable is limited by how well that variable can be predicted. A variable can be highly predictive at the individual level, but if it cannot be accurately predicted at a county-by-stratum level, it could hurt the predictive ability of the model.

Obtaining Confidence Intervals and Bootstrapping

After generating small area estimates that reflect each small area's demographics, the next step is to obtain confidence intervals, which capture uncertainty. The researchers at the UofSC accomplished this through a Monte Carlo–based parametric bootstrapping approach (Tofighi & MacKinnon, 2016), repeated 10,000 times. At every iteration, new random and fixed effects were randomly generated based on their distributions, as estimated by the small area model. Using these randomly generated coefficients, prevalence predictions were obtained for all strata. Then, poststratification was applied to the new strata-level prevalence predictions (the necessary population counts do not change with each iteration), giving new small area estimates. For each small area, the 95% confidence interval (CI) was either given by the 2.5th

and 97.5th percentiles of the bootstrap sample or derived from the sample's standard deviation, and the point estimates were given either by the medians or the means, respectively.

SAE Utility in Action: Examples of Predicted Estimates Using the National Survey of Children's Health

The following section provides more detailed examples of how the research team at the UofSC used SAE to estimate Tourette Syndrome (TS) and overweight/obesity prevalence rates at the county-, state-, and Census regional-levels from the public- and restricted-use 2016-2020 National Survey of Children's Health, respectively. These examples highlight specific ways in which poststratification can be achieved and the utility of SAE in multiple contexts. In both examples, the research team had already merged the NSCH data with publicly available data, used the same variable selection approach, and utilized bootstrapping to obtain confidence intervals.

Tourette Syndrome (TS): SAE with Publicly Available Data

TS Background

Tourette Syndrome is a neurological disorder characterized by repetitive movements and/or unwanted vocal sounds, with 0.3% of U.S. children having a diagnosis based on NSCH data (Bitsko et al., 2022). Given the lower estimate of diagnosed TS, smaller area estimates can be particularly helpful in identifying specific areas in which diagnostic and treatment resources may be most beneficial. To address this need, the research team at the UofSC generated state-level prevalence estimates of TS using child- and area-level predictors. This example illustrates how to approach poststratification using publicly available data. Although states are not small areas per se, state-level prevalence estimates of TS can still be considered small area estimates (SAE) due to the low prevalence of diagnosed TS. Nationally, three out of every 1000 children have TS (Bitsko et al., 2022) and the sample size for each year of the NSCH is approximately 1000 children per state. As a result, producing estimates at the state-level is a challenging task.

TS Model

The research team at the UofSC applied SAE, predicting state-level prevalence estimates of TS from child- and area-level information. Specifically, the model included child-level information on age, biological sex, the interaction between age and sex, and year of participation in the NSCH survey, as well as area-level information, such as the number pediatricians per 100,000 state residents, the proportion of state residents with "some college" education, and the interaction between these two area-level variables. This model also included a random intercept for each state.

This model did not include any spatial effects because the variable selection process indicated that the CV RMSPE was minimized in the absence of spatial components. The model building process was iterative, wherein previous models with different predictors were examined (e.g., neurodevelopmental disorder status), but these models were ultimately discarded because they produced results with too much uncertainty (e.g., wide confidence intervals).

TS Poststratification

As with the general SAE approach outlined previously, the regression results provided TS estimates for each child-level stratum (i.e., survey participation year, sex, and age). To make a single TS estimate, the researchers needed to perform poststratification. In this scenario, there were 120 strata (120 = 5 survey years * 2 sexes * 12 ages), so population counts for each stratum needed to be obtained. At the *state-level*, population counts for male and female children, as well for ages and survey year (number of children in each state in 2016, 2017, 2018, 2019, and 2020) were accessible through the Census, so the researchers used that publicly available data.

Then, using these state-level population counts, as well as the TS estimated prevalence per stratum, the researchers were able to calculate a weighted average per state, which represented the state-level TS prevalence. To estimate variability, researchers used a Monte Carlo–based parametric bootstrapping approach (Tofighi & MacKinnon, 2016) with 10,000 iterations, such that the 95% confidence intervals were formed by using the standard deviations of the bootstrap samples, and the point estimates were given by the means.

Benchmarking. To obtain a *national* TS estimate, the researchers took a final weighted average, considering the number of children in each state (excluding Hawaii and District of Columbia due to missingness in area-level variables for these two states). Through this process, the researchers at the UofSC obtained a national TS estimate of 0.37%. However, research collaborators at the CDC have consistently obtained a national estimate of 0.31% over the past five years using standard approaches to analyzing NSCH data. This is a meaningful difference given the relatively low prevalence of diagnosed TS. To account for this difference, the researchers at the UofSC used benchmarking, which is a specific type of poststratification. First, they created a benchmarking weight (i.e., the CDC national TS estimate divided by the UofSC national TS estimate; $0.0031/0.0037 \approx 0.84$), and then they multiplied each bootstrap generated state-level TS prevalence estimate by the benchmarking weight. In doing so, when the researchers at the UofSC generated an updated national TS prevalence estimate, their national estimate matched that of the CDC (i.e., 0.31%), instilling more confidence in their updated state-level estimates.

TS Results: State-Level Estimated Prevalence Estimates

Figure 1 shows the state-level results from the SAE analysis with benchmarking. Prevalence estimates ranged from 0.42% to 0.22%. The highest estimates were located in Pennsylvania (0.42%) and New Jersey (0.41%), as well as in Vermont, West Virginia, and Maryland (all 0.39%), while the lowest estimates were found in North Dakota (0.22%), Idaho (0.23%), Wyoming (0.24%), and Utah (0.24%). While there is only a 0.20% difference between the states with the highest and lowest estimates, states with higher TS estimates have prevalence estimates up to 35% *higher* than the estimated national estimate. Likewise, states with lower TS estimates have prevalence estimates up to 29% *lower* than the national estimate, suggesting that small state-level differences may still be meaningful from a targeted policy implementation perspective. However, it is worth acknowledging that the 95% confidence intervals (CIs) for states with the highest estimates and with the lowest estimates overlap. While this overlap suggests that there

is uncertainty in the degree to which state estimates are statistically different from one another, the estimates can still help guide implementation decisions.

Overweight and Obesity: SAE With and Without Publicly Available Data and Simplifying Assumptions[4]

Overweight/Obesity Background

Overweight and obesity, particularly in children, are a major public health concern, as they have been associated with a host of physical and mental challenges, including decreased life

Figure 1: State-Level Tourette Syndrome Estimates from 2016-2020 NSCH Dataset

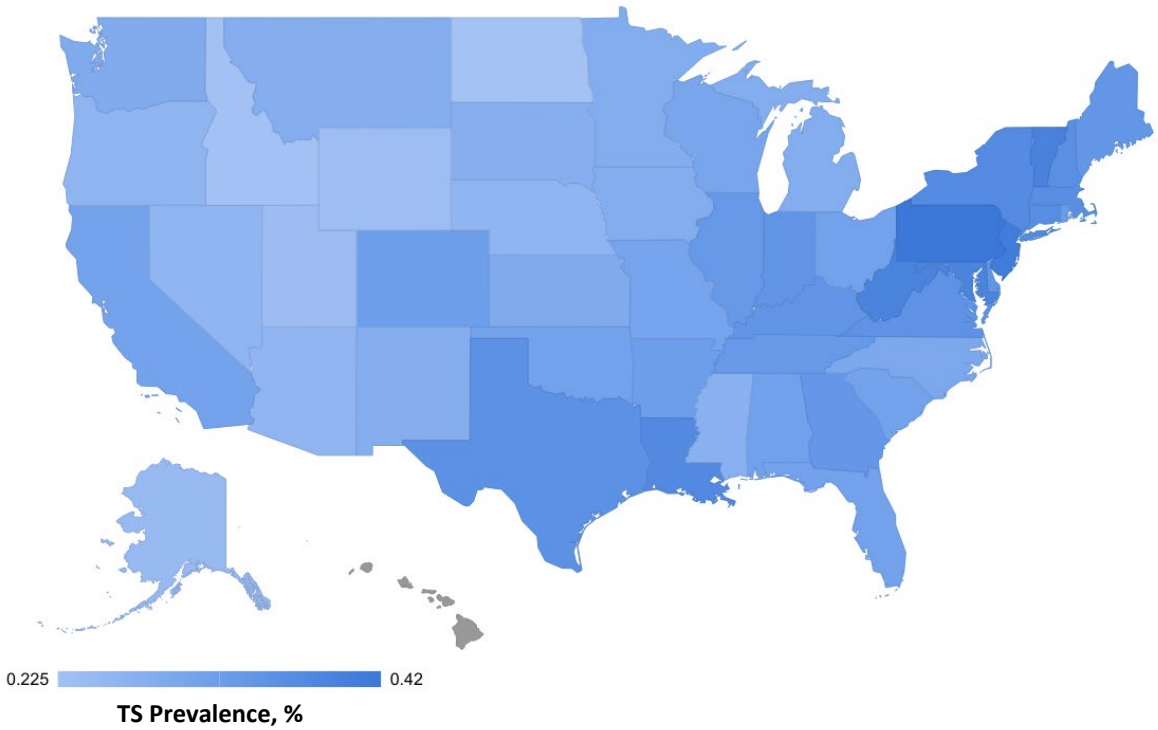


Figure 1. Choropleth map displaying range of TS prevalence estimates in the U.S. Darker blue represents higher estimates, while lighter blue represents lower estimates. State-level estimates were not available for Hawaii or the District of Columbia.

expectancy (Ebbeling et al., 2002; Halfon et al., 2013; US Preventive Services Task Force et al., 2017). Almost 20% of children are obese (Bryan et al., 2021), but this estimate is not constant across geographical locations or demographic characteristics (Y. Wang et al., 2008; Y. C. Wang et al., 2011),pointing to the need to have accurate, up-to-date precise county-level estimates. This precision can help ensure that policies and programs are targeting the areas in most need of supports. Consequently, the research team at the UofSC predicted both county-level and

[4] This SAE analysis was first published in the *American Journal of Epidemiology*. See Zgodic et al., 2021 for complete details.

regional-level prevalence estimates of overweight/obesity using child-level and area-level predictors (Zgodic et al., 2021). This example illustrates how to approach poststratification when publicly available data does not provide all necessary population counts.

Overweight/Obesity Model

The research team at the UofSC applied SAE via a multilevel mixed effects logistic regression model with a county-level intrinsic conditional autoregressive random intercept term (ICAR; Banerjee et al., 2004; Cressie, 1993), predicting county-level prevalence estimates of overweight/obesity from child- and area-level information. Specifically, the model included information on child race/ethnicity, age, sex, and highest parental educational attainment, as well as area-level information, such as regional Census division, county percentage of children in single parent household, the number primary care providers per 100,000 county residents, county adult obesity rate, presence of a state school wellness policy, county-level population weighted walkability index, and county rurality status. This model included child-level and area-level fixed effects, as well as an ICAR random intercept term, accounting for spatial dependence between neighboring counties.

Overweight/Obesity Poststratification

The regression model provided overweight/obesity estimates for all possible combinations of child-level predictors. In this example, there were 256 strata per county (256 = 4 races/ethnicities* 2 sexes * 8 ages * 4 highest parental educational attainments). As in the approach employed for the TS model, the researchers needed to obtain population counts to estimate a single, representative prevalence rate per county. At the *county-level*, population counts for race/ethnicity, sex, and age were accessible through the Census, so the researchers used that publicly available data.

However, the population counts for children's highest parental educational attainment were not publicly available. As a result, several simplifying assumptions using publicly available Census data across a multi-step process were made in order to calculate these population counts (Zgodic et al., 2021). First, it was assumed that a parent's education was independent of their child's age and gender, but not independent of their child's race. As such, the researchers sought to determine the probability of parental educational attainment given race. First, using Census population counts, the researchers calculated the county-to-state ratio of adults with a given educational level conditional on a particular race. Since the Census only provided the prevalence of adult (not parental) education given race, the UofSC researchers assumed that this ratio was the same for parents sampled in the NSCH data. Next, the researchers used the NSCH data to fit a mixed effects multinomial regression model that estimated the state-level prevalence of parental education given child race/ethnicity and Census regional division. Finally, using this model's predicted education level prevalence and the ratio of county-to-state adults/parents with a given educational level, the researchers calculated the county-level prevalence of parents with a certain educational attainment given their child's race.

Using Census county-level population counts, the estimated education stratum counts, and the overweight/obesity probabilities per stratum, the researchers were then able to calculate a

weighted average per county. Finally, to estimate variability, researchers used a Monte Carlo–based parametric bootstrapping approach (Tofighi & MacKinnon, 2016) with 10,000 iterations, such that the 95% confidence intervals were given by the 2.5^{th} and 97.5^{th} percentiles of the bootstrap samples, and the point estimates were given by the medians.

Overweight/Obesity Results: Where SAE Works Well

Nationally, overweight/obesity estimates were estimated to be 30.7%; regionally, the East South Central regional census division had the highest estimate (36.3%) of overweight/obesity, whereas New England had the lowest estimate (27.8%). County-level differences emerged as well. Specifically, 57.5% of counties in the Southeast (i.e., West South Central, South Atlantic, and East South-Central divisions) had overweight/obesity estimates greater than 33%, while only 21.8% of counties in the Midwest and on the West coast (i.e., West North Central, Mountain, and Pacific divisions) had overweight/obesity estimates greater than 33%. See Zgodic et al., 2021 for complete results.

In terms of certainty, the width of the confidence intervals revealed meaningful patterns. Generally, narrow confidence intervals suggest more certain estimates, whereas wide confidence intervals indicate less certainty. Interestingly, there was considerable *uncertainty* (95% CI width = 66.9%) in the estimate for Los Angeles County, CA (42.7%, 95% CI: 11.9%, 78.8%) but much more *certainty* (95% CI width = 26.6%) in the estimate for Dallas County, TX (44.4%, 95% CI: 31.3%, 57.9%) despite both being large metropolitan counties and with similar overweight/obesity estimates (~43-44%). The discrepancy in uncertainty is likely due to fixed effects via the different demographics of the areas. Specifically, Los Angeles has higher estimates of particular underrepresented race/ethnic groups by parental education attainment. These combinations were rare in other counties throughout the U.S. Consequently, the fixed effect model coefficients for those strata are estimated with more error. Conversely, Dallas has sociodemographic characteristics that are more similar to the rest of the U.S. As a result, the model coefficients for those strata are estimated with less error. Generally, these results suggest that SAE is not an equally reliable method for all communities; more specifically, these results indicate that SAE provides estimates with narrow confidence intervals for more homogeneous, rather than heterogeneous, counties.

Summary

Small area estimation (SAE) is a powerful tool with many strengths. Primarily, it estimates prevalence estimates in areas with small sample sizes by borrowing information from areas with larger sample sizes. Moreover, it is also a flexible technique. SAE can employ individual-level, as well as area-level, predictors for data that is publicly available (e.g., Tourette Syndrome), for data that requires simplifying assumptions (e.g., overweight/obesity), and for data that relies on other SAE models (e.g., autism spectrum disorder; Bradshaw et al., under review). SAE can also incorporate spatial information using intrinsic conditional autoregressive random intercept terms (ICAR; Banerjee et al., 2004; Cressie, 1993).

However, despite its strengths, SAE is not without limitations. Namely, it is not a ubiquitously useful method. In areas with sociodemographic homogeneity, it provides more certain estimates, but in areas with sociodemographic heterogeneity, it generates less certain estimates. As SAE is a method that can identify which communities most need support, certainty is important to consider. Stakeholders would want to prioritize interventions in areas where one is more confident of a community's needs. Moving forward, researchers should continue to utilize SAE because it is well suited to provide community-specific information regarding where interventions, policies, and resources are needed, but they should carefully monitor the certainty of their results.

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